

Macroeconomic regime identification using a two-step approach with independent component analysis and hidden Markov models

Barenda Rundle*

Institute of Finance and Technology, University College London
and

Francesca Medda

Institute of Finance and Technology, University College London

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Abstract

Hidden Markov models are often used to identify different regimes. However, in a multivariate setting, correlations between variables may skew the results, leading to potentially flawed analyses. This paper proposes a two-step approach to better identify hidden regimes in macroeconomic time series. In the first step, independent components are extracted from nine macroeconomic time series using second order blind identification (SOBI). In the second step, the independent components are used in a hidden Markov model to identify macroeconomic regimes. The results from the two-step process show increased regime persistence compared with a pure hidden Markov model, suggesting clearer identification of regimes when dealing with correlated time series. The paper also introduces two new measures of the quality of regime classification.

Keywords: Macroeconomic regimes; Hidden Markov models; Independent component analysis

JEL Classification: C00, C38, E00

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1 Introduction

The financial crises of 2008 has put macroeconomic modelling back in the spotlight. Traditional dynamic stochastic general equilibrium models (DSGE) seemed to, at best, fail in foreseeing the crisis and at worst, contributed to the crisis by underestimating the risks. Critics of DSGE-models argued that the models were flawed for various reasons, such as being based on unappealing assumptions and methods of estimation among others (Blanchard, 2016), or being self-referencing and inward-looking, unable to describe the non-linear dynamics of economic fluctuations (Buiter, 2009). As a result, there has been a call for approaching macroeconomic modelling from different perspectives (Blanchard, 2018).

In light of these considerations, we consider two aspects to macroeconomic modelling. Firstly, we argue that the cyclical or non-linearity of the system is due to the presence of different macroeconomic regimes. Secondly, we propose that the observed data is a mixture of several latent economic influences that may each individually influence the broader economy, but together cause shifts in equilibrium regimes. Our proposed model is a combination of independent component analysis and Markov-switching in order to test these two aspects.

The use of regime-switching models to explain the non-linearity in the market is not new. Hamilton (1989) first used a regime-switching approach to model the non-linearity of US GNP growth. Since then, an extensive body of research has been done on identifying different macroeconomic and/or market regimes. Evans and Wachtel (1993) investigated the presence of regimes in inflation data, while Engel (1994) and Frömmel et al. (2005) applied regime-switching models to exchange rates. Hamilton and Susmel (1994) and Cai and Cal (1994) added conditional heteroscedasticity to the regime switching approach to model the volatility in stock returns and interest rates, respectively. In most cases, the regime switching model was fitted to single series, with the aim of modelling the underlying regimes in those timeseries. Hamilton et al. (2016, Section 3) provides a framework of how regime-shifting behaviour can be represented in theoretical models of the economy.

In the 2000s, research shifted to the regime-dependent effects of macroeconomic variables on stock returns (Chang, 2009; Chen, 2007, 2008). Henkel et al. (2011) used a regime switching vector autoregression (RS-VAR) model and showed that the ability of dividend yield and term structure variables to predict stock returns increased during recessions.

Other work focused on optimal asset allocation strategies under a regime-switching model (Ang and Bekaert, 2003; Guidolin et al., 2005; Guidolin, 2011). Ang and Timmermann (2012) provides a detailed review of the literature focusing on regime changes and financial markets. More recently, higher-order hidden Markov models were used in an asset allocation setting (Zhu et al., 2017) to account for longer memories in financial markets.

However, as far as we could determine, little research has been done on regime shifts that focussed on the broad, multivariate macroeconomic environment. Baele et al. (2015) introduced a new New-Keynesian model that accommodates regime-switching behaviour, and used four regime variables that captures the shifts in monetary policy, as well as the volatility in supply shocks, demand shocks and monetary policy shocks. This paper investigates the presence of shifts in the broader economy outside the framework of an established economic model, using a wide range of economic indicators. Some of these variables are correlated and/or temporally correlated, which could lead to an economic indicator displaying characteristics of a regime change that is caused by another variable. In a multivariate hidden Markov model, this could result in more macroeconomic regimes being identified than what is actually occurring. We therefore propose to preprocess economic indicators using independent component analysis to obtain independent signals that could influence regime shifts.

This paper compares the results from two approaches to regime detection. In the first approach, regimes are modelled using a hidden Markov model (Rabiner, 1989) on nine different economic indicators covering interest rates, inflation, money supply, production, exchange rates and stock prices. In the second approach, we first perform independent component analysis (Hyvärinen et al., 2001) to extract mutually independent components from often-correlated macroeconomic indicators. Since the underlying influences may have temporal dependencies, we used second order blind identification (SOBI) (Belouchrani et al., 1997) to extract independent components, before modelling the hidden regimes using a hidden Markov model, using the components as input.

Our results empirically confirm the model first introduced by Goodwin (1951) which argued that the model economy evolves smoothly with multiple equilibria. In an extension of this work, Cooper (1994) argued that agents' inferences about the actions of other agents move slowly, and hence the next equilibrium will be relatively close to the current

equilibrium. This is further extended and combined with regime-switching in Chamley (1999).

The paper is organised as follows. The next section provides an overview of hidden Markov models and independent component analysis, and describe the SOBI algorithm used to extract independent components, and introduces the model used in this paper. Section 3 describes the data used in the analysis and includes a short discussion of the observed correlations and cross-correlations as well as a traditional vector autoregressive model of the variables. Section 4 contains the results from hidden Markov models on the untransformed data, both univariate and multivariate. Section 5 contains the results from the independent component analysis and the hidden Markov models based on this input. Section 6 concludes.

2 The model

2.1 Hidden Markov Model

This section provides a brief overview of hidden Markov models (HMM). A comprehensive description is found in Rabiner (1989) and in Cappé et al. (2005).

The basis of a HMM is a Markov chain, where the system may be described as being in one of a set of K states, $\mathbf{S} = S_1, S_2, \dots, S_K$ at time $t = 1, \dots, T$. For a first-order Markov chain, the probability of being in a state q_t only depends on the previous state. The Markov chain is specified by the transition probability matrix, $\mathbf{A}$, where a_{ij} is the probability of moving from state i to state j . The initial state probabilities for each state are given by $\boldsymbol{\pi} = \pi_1, \dots, \pi_K$ with $\sum_k^K \pi_k = 1$.

For HMMs the states are hidden and each state or regime generates multivariate observations $\mathbf{y}_t$ at time t which are consistent with the observation model under that particular regime. The observation model has a set of parameters, $\boldsymbol{\theta} = \{\theta_1, \dots, \theta_k, \dots, \theta_K\}$.

The parameters of an HMM can be concatenated into a parameter set $\Theta = \{\mathbf{A}, \boldsymbol{\pi}, \boldsymbol{\theta}\}$. The joint log-likelihood of a state sequence and an observation model is given by:

$$\log p(\mathbf{y}, \mathbf{q} | \Theta) = \log p(q_1 | \boldsymbol{\pi}) + \sum_{t=2}^T \log p(q_t | q_{t-1}) + \sum_{t=1}^T \log p(\mathbf{y}_t | q_t, \theta_{q_t}) \quad (1)$$

The three terms refer to the log-likelihood of the initial probabilities, transition matrix and observation model, respectively.

The hidden states and the observation model under each state can be estimated simultaneously using an expectation-maximisation (EM) algorithm (Baum et al., 1970; Dempster et al., 1977). The EM algorithm finds the maximum likelihood estimates $\hat{\Theta}$ by iterating between an **expectation** step, where the posterior probability $p(\mathbf{q}|\mathbf{Y}, \Theta)$ is calculated, and a **maximisation** step, where the parameters $\hat{\Theta}$ are updated so that it maximises (1). The expectation and maximisation steps are repeated until convergence.

In this analysis, $\mathbf{y}_t$ are the independent components extracted from mixed signals. We assumed that the observation model is a Gaussian probability distribution with a regime-specific mean and covariance with $\theta_k = \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}$.

2.2 Independent component analysis

The second method that forms the basis of our model is independent component analysis. As above, we provide a brief overview of the method that we used in this analysis. A more comprehensive overview of the various methods is available in Hyvärinen et al. (2001).

Blind source separation (BSS) methods attempts to extract the original sources from a set of mixed signals. No information is available on either the original source signals or on the mixing process. One well-known example is the cocktail party problem, where the voices of many people speaking simultaneously are recorded by a number of microphones. The microphones will record a mixture of the voices and the objective of BSS is to extract the original voices from the mixtures. Independent component analysis (ICA) is one BSS method that have showed success in extracting independent signals from a range of different applications including finance and banking (Back and Weigend, 1997; Oja et al., 2000; Lin and Chiu, 2013).

Contrary to principal component analysis (PCA), which extract *uncorrelated* components from a number of series, the aim of ICA is to extract statistically *independent* components from a set of observed, mixed signals.

Consider a set of m observed random variables, $\mathbf{x}(t) = (x_1(t), \dots, x_m(t))^T$, which is assumed to be a linear mixture of l independent sources $\mathbf{s}(t)$, where $l \leq m$. This linear mixture can be represented by

$$\mathbf{x}(t) = \mathbf{M}\mathbf{s}(t) + \mathbf{v}(t) \quad (2)$$

$$\text{where } p(\mathbf{s}(t)) = \prod_{i=1}^l p_i(s_i(t)) \quad (3)$$

where $\mathbf{M}$ is a $m \times l$ dimensional matrix, and $\mathbf{v}$ is additive white noise.

The objective of ICA is to find the matrix $\mathbf{W}$ that could invert the mixed information to yield the original sources:

$$\mathbf{y}(t) = \mathbf{W}\mathbf{x}(t) \quad (4)$$

$$\text{such that } p(\mathbf{y}(t)) = \prod_{i=1}^L p_i(y_i(t))$$

The solution to 4 is not unique and many different algorithms have been introduced to estimate the mixing matrix and independent components. Of these, the most commonly used algorithms are the joint approximate diagonalisation of eigenmatrices (JADE) (Cardoso and Souloumiac, 1993), Infomax (Bell and Sejnowski, 1995), FastICA (Hyvärinen and Oja, 1997) and second order blind identification (SOBI) (Belouchrani et al., 1997). Since the macroeconomic indicators are time-series that could have different time dependencies, SOBI was used in this analysis.

2.2.1 Second order blind identification (SOBI)

The SOBI algorithm uses joint diagonalisation of several covariance matrices to extract independent components. The algorithm consists of the following steps (see Belouchrani et al. (1997) for the derivation).

1. Estimate the sample covariance $\hat{\mathbf{R}}(0)$ from T data samples, and find the n largest eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ and associated eigenvectors $\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n$ of $\hat{\mathbf{R}}(0)$;
2. Determine the whitening matrix:

$$\hat{\mathbf{W}} = [(\lambda_1 - \hat{\sigma}^2)^{(-1/2)}\mathbf{h}_1, \dots, (\lambda_n - \hat{\sigma}^2)^{(-1/2)}\mathbf{h}_n]^H \quad (5)$$

where H indicates the Hermitian transpose of the matrix and $\hat{\sigma}$ is an estimate of the noise variance, which can be calculated as the average of the smallest eigenvalues $\lambda_{n+1}, \dots, \lambda_m$;

3. Calculate the whitened signals, $\mathbf{z}(t) = \hat{\mathbf{W}}\mathbf{x}(t)$;
4. For a fixed set of time lags, $\tau \in \{\tau_j | j = 1, \dots, K\}$, compute the sample covariances matrix $\hat{\mathbf{R}}(\tau)$;
5. Determine the unitary matrix $\hat{\mathbf{U}}$ that jointly diagonalises the set:

$$\{\hat{\mathbf{R}}(\tau_j) | j = 1, \dots, K\}; \quad (6)$$

6. Estimate the sources as:

$$\hat{\mathbf{s}}(t) = \hat{\mathbf{U}}^H \hat{\mathbf{W}} \mathbf{x}(t) \quad (7)$$

and the mixing matrix as $\hat{\mathbf{A}} = \hat{\mathbf{W}}^\# \hat{\mathbf{U}}$ where $\#$ denotes the Moore-Penrose pseudo-inverse.

The SOBI algorithm assumes that the extracted components are weakly (second-order) stationary with different time dependence structures. Choi and Cichocki (2000) introduced an extension to SOBI that allows the variances of the components to be non-stationary. In this paper, we model the non-stationarity in components explicitly using multivariate hidden Markov models, in which both the mean and the variance of the components can be non-stationary.

2.3 ICA-HMM model

The motivation for a combined ICA-HMM model is the need to analyse the macroeconomic environment which, in turn, is the aggregation of multiple related economic indicators.

Economic indicators and the macroeconomic environment as a whole are known to be non-stationary. However, at the same time the underlying variables contributing to the macroeconomy are interdependent, and could therefore be considered a range of mixed economic influences. For instance, the exchange rate could be considered a mixture of the interest rate differentials between different countries, inflation, balance of trade and the economic strength of a country. Similarly, interest rates could be considered a mixture of inflation and the monetary or fiscal policies. It is our view that by extracting the independent sources or influences from the mixed signals using ICA, a better understanding of the regime-shifting process can be obtained.

Formally, consider m economic time-series $\mathbf{x}(t) = (x_1(t), \dots, x_m(t))_T$. The model uses the SOBI algorithm from Section 2.2.1 to extract independent economic influences, **SOBI**(t) where each series within the system is higher-order independent from each other. These independent influences are then used in a HMM to model the shifts in regimes. The single best regime or state-path can be decoded from the HMM using the Viterbi algorithm (Viterbi, 1967; Forney, 1973).

2.4 Model selection

Random values are often used to initialise the parameters of the HMM and fine-tuned using the expectation maximisation algorithm described in Section 2.1. However, depending on the starting values, the log-likelihood could converge at a local maxima with suboptimal estimates of the parameters. It is also possible that there are a number of local maxima that are all equally good but provide different estimates of the parameters and hence different regime classifications. It is therefore advisable to train the model with several randomisations and select the "best" model from the possible candidates.

Furthermore, while some parameters (transition matrix, initial probabilities and parameters of the observation model) of a specific HMM can be estimated using the EM algorithm described above, the number of regimes to include and the observation probability distribution has to be decided beforehand. The log-likelihood of the data under each model could be used, but could lead to overfitting as increasing model complexity could increasingly fit the data.

The need for a consistent measure of model selection is therefore required. The Akaike and Bayesian information criteria (AIC and BIC) are often used to decide between competing models. These criteria balances higher log-likelihood with the number of parameters to estimate and hence reduce the probability of overfitting.

Specifically, the Bayesian information criteria (BIC) (Schwarz, 1978) penalises model complexity and is calculated as:

$$\text{BIC} = np \ln(n) - 2 \ln(L(\hat{\Theta})) \quad (8)$$

where

$L(\hat{\Theta})$ = maximum likelihood of the model under model parameters $\hat{\Theta}$

np = the number of parameters estimated by the model

n = the number of observations

The argument for using BIC rather than AIC is that as $n \rightarrow \infty$, the BIC is more likely to select the model that generated the data. In this analysis, we used the BIC to select the model.

In addition, the transition matrix can provide an indication of how often regimes change. We define the regime persistence measure as:

$$RP = \frac{\sum_k^K a_{kk}}{K} \quad (9)$$

where K is the number of regimes and a_{kk} is the probability of remaining in regime k (that is, the diagonal of the transition probability matrix, $\mathbf{A}$).

The RP of a very persistent HMM will be close to 1, while a HMM where the regimes change at every step will have a RP of zero. In this article, we express the RP as a percentage.

Finally, Ang and Bekaert (2002) considered the quality of regime classification as the ambiguity of each regime and introduced the regime classification measure (RCM), given by:

$$RCM = 100K^2 \frac{1}{T} \sum_{t=1}^T \left(\prod_{k=1}^K p_{t,k} \right) \quad (10)$$

where K is the number of regimes, T in the number of observations and $p_{t,k}$ is the probability of regime k at time t .

In this definition, regime inference is weak when the model cannot successfully distinguish between regimes, in other words when the probability of two (or more) states are similar. A higher quality classification is association with a lower RCM. However, if there are more than two regimes, the probability of one regime could be zero, with the remaining probabilities fairly equal. The above measure will in this case not penalise the classification for the relative ambiguity in classification. Our modified regime classification measure (MRCM) caters for this situation and is given by:

$$\text{MRCM} = \frac{100}{T(1 - \frac{1}{K})^K} \sum_{t=1}^T \left(\prod_{k=1}^K (1 - p_{t,k}) \right) \quad (11)$$

Consistent with Ang and Bekaert (2002) we express this measure as a percentage between 0% (no ambiguity) and 100% (no certainty), with a lower measure preferred.

3 Data

We illustrate this model using UK economic data. Nine monthly economic time series were obtained from the Bank of England (Thomas and Dimsdale, 2017) covering various aspects of the macroeconomic environment that could affect monetary policy and the broad investment horizon. These variables and their transformations are listed in Table 1. Data between January 1960 and December 2016 were used in the analysis and based on Augmented Dicky-Fuller tests, these series are stationary. The data was standardised to remove scaling inequalities.

Table 1: Variables considered in the analysis and their transformation

Variable	Abbreviation	Transformation
Bank deposit rate	Bank_deposit	First difference
Medium-term bond yield	MT-yield	First difference
Corporate bonds spreads	CB_spreads	First difference
Index of Industrial Production	IP	First difference of log
Stock index	Stocks	First difference of log
\$/£ exchange rate	Dollar_pound	First difference of log
Producer price index	PPI	First difference of log
Notes and coin in circulation	M1	First difference of log
Aggregate Credit	Credit	Second difference of log

The time series of the variables are shown in Figure 1.

From the plots, two observations can be made:

1. **Relationships between variables:** Some variables, for instance the bank deposit rate and the medium-term yield, seem to be correlated and temporally cross-correlated

(correlated over different leads and lags). In Section 3.1, we modelled the relationship between variables using a vector autoregression model.

2. **Different regimes:** Although all the time series were standardised, there appear to be periods of increased volatility and a shift in the location. For instance, the a number of indicators displayed increased volatility in 1970s, which was characterised by the oil crises in 1973 and 1979, leading to soaring inflation, which is evident in above-average PPI during that period. Although less pronounced, several time series reflect the credit crisis in 2008, while individual indicators display periods of higher volatility or a shift in mean for a sustained period, for instance, corporate bond spreads in 1985-1995. We model the regimes based on raw variables in Section 4.1.

3.1 Vector autoregression

In this section, we consider the relationship between the economic series. Firstly, the Pearson correlations of the raw series are shown in Figure 2. There is generally low correlation between the series, with the exception of the relationship between long-term and short-term interest rates, and the negative correlation between stock returns and bond returns. However, there may be relationships between the variables at various leads and lags. For instance, the temporal cross-correlation between industrial production and inflation, suggests that industrial production (IP) is negatively impacted by higher inflation (PPI) four to 12 months later (Figure 3).

As a second step in understanding the relationships between the various variables, we fitted a vector autoregression (VAR) model, defined for m variables $\mathbf{x}(t) = (x_1(t), \dots, x_m(t))^T$ as:

$$\mathbf{x}(t) = A_1 \mathbf{x}_{t-1} + \dots + A_p \mathbf{x}_{t-p} + \mathbf{u}_t \quad (12)$$

where A_i are $m \times m$ coefficient matrices for $i = 1, \dots, p$ and $\mathbf{u}_t$ is an m -dimensional white noise process with $\mathbb{E}(\mathbf{u}_t) = \mathbf{0}$ and time-invariant covariance matrix $\mathbb{E}(\mathbf{u}_t \mathbf{u}_t^T) = \Sigma_{\mathbf{u}}$.

In this analysis, the number of lags to include for each variable was based on the Bayesian information criteria (BIC) as discussed in Section 2.4. The system of equations from the VAR based on a 95% confidence level is given by:

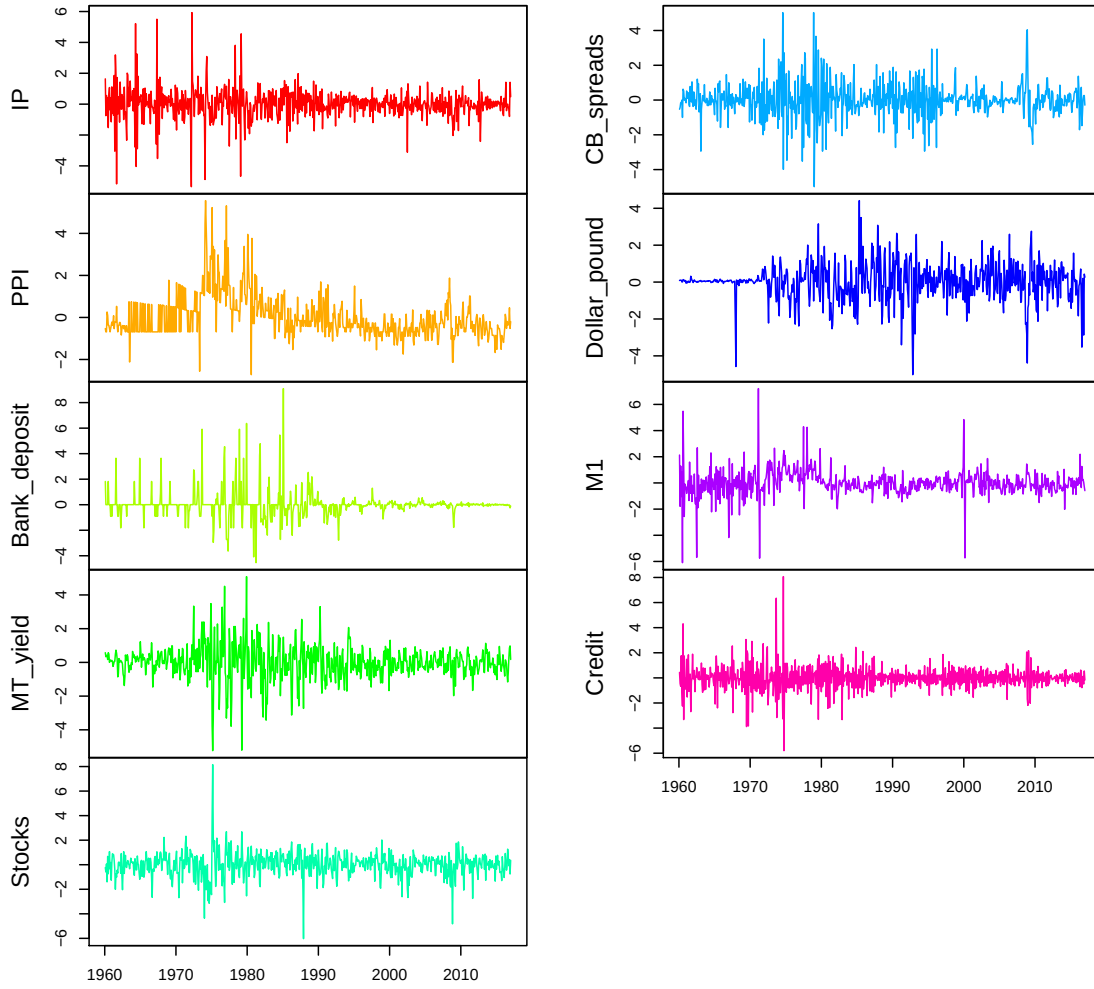


Figure 1: Time series of transformed variables

$$\begin{aligned}
 IP_t &= -0.33 IP_{t-1} + 0.11 MT_yield_{t-1} \\
 PPI_t &= 0.55 PPI_{t-1} - 0.07 Dollar_pound_{t-1} \\
 Bank_deposit_t &= 0.28 MT_yield_{t-1} \\
 MT_yield_t &= 0.37 MT_yield_{t-1} \\
 Stocks_t &= -0.13 MT_yield_{t-1} + 0.24 Stocks_{t-1} \\
 CB\ Spreads_t &= -0.28 Stocks_{t-1} - 0.21 CB_Spreads_{t-1} - 0.10 Dollar_pound_{t-1} \\
 Dollar_pound_t &= 0.13 Bank_deposit_{t-1} + 0.35 Dollar_pound_{t-1} \\
 M1_t &= -0.10 IP_{t-1} + 0.15 PPI_{t-1} - 0.08 Dollar_pound_{t-1} \\
 Credit_t &= -0.06 M1_{t-1} - 0.53 Credit_{t-1}
 \end{aligned} \tag{13}$$

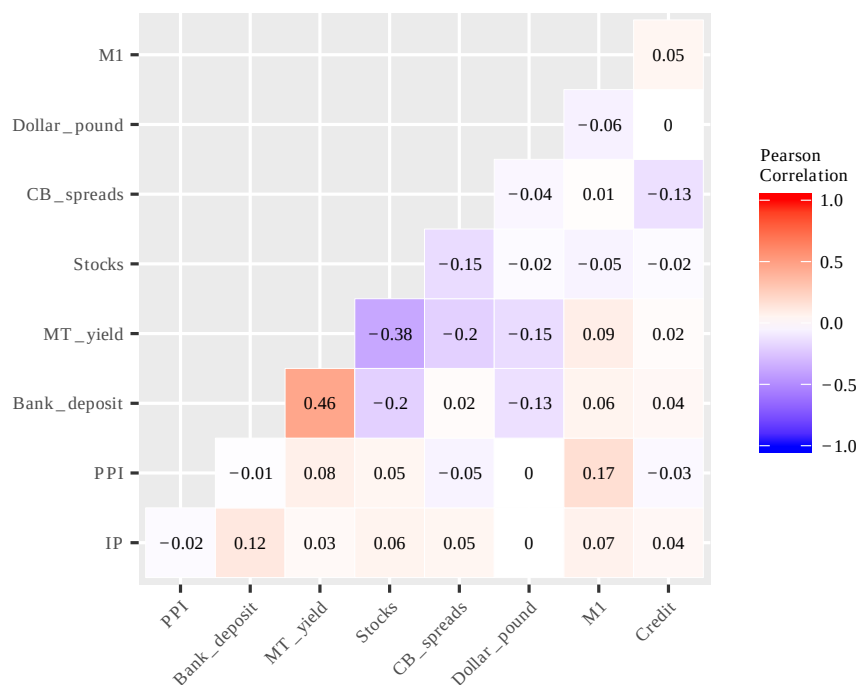


Figure 2: Correlations between economic indicators

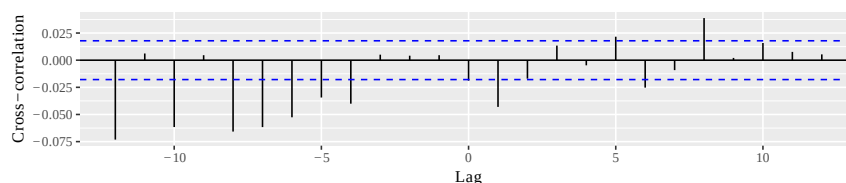


Figure 3: Cross-correlations between PPI and IP at different lags of PPI

Not surprisingly given the cross-correlations, most of these variables have impact on other variables in the system. Some of the relationships are fairly intuitive: for instance, it is conceivable that there is a relationship between PPI and the exchange rate; similarly, that there is a negative relationship between stock returns and medium-term bond yields. Furthermore, it is interesting to note that in three of the variables, there are strong indications of short-term mean-reversion: IP, CB spreads and Credit all have *negative* coefficients with their respective lagged values.

It may be useful to identify the independent driving influences that may affect the economy as a whole, without the interference of other secondary impacts. For instance,

there is likely to be a "effect" that is purely related to interest rates, without the additional and indirect influence of the exchange rate. Similarly, there is likely to be an effect purely related to equity risks, without the cross-impact of lagged interest rates. By analysis these "pure" and independent effects, we could get clearer indication of changes in the overall economic environment.

4 Results: Hidden Markov Models on untransformed variables

In this section, we fit Hidden Markov Models (HMM) on the raw, untransformed variables for both individual indicators and the multivariate case.

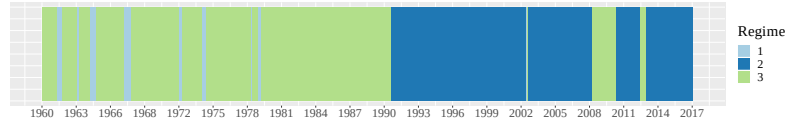
4.1 HMMs on individual economic indicators

Hidden Markov models were fitted to each individual economic indicator and the Viterbi algorithm was used to determine the most likely regime over time.

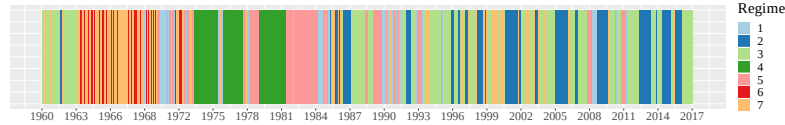
We provide below a short description of the regimes for each indicator. For each variable, the hidden Markov model was estimated assuming a mixture of Gaussian distributions. The plots of the different regimes identified for each of the economic indicators appear in Figure 4.

4.1.1 Industrial production

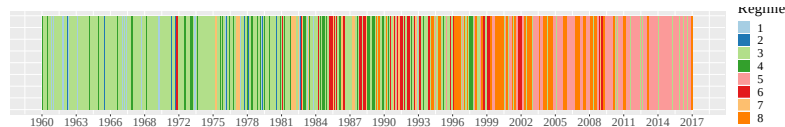
In this case, three regimes were identified (Figure 4a). The descriptive statistics of the standardised industrial production in each regime is shown in Table 2. There seems to be two equilibrium regimes, one from the start of the period until mid-1990 (regime 2), where industrial production was below the long-term average, coinciding with the Thatcher-era. The second equilibrium regime was from mid-1990 to the end of the study period, where industrial production was marginally above average. The first equilibrium regime was interrupted by regime 1, mostly in the 1960s, which has relatively low persistence and associated with higher-than-average industrial production, with significantly higher levels of volatility. The second equilibrium regime was interrupted by the second, lower-growth regime (regime 2). Of particular interest is that the regime changed to the lower-growth



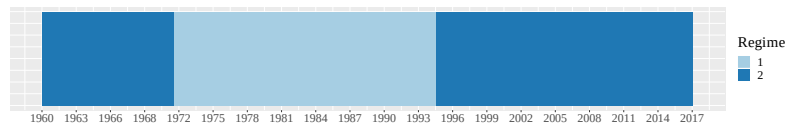
(a) Regimes based on industrial production



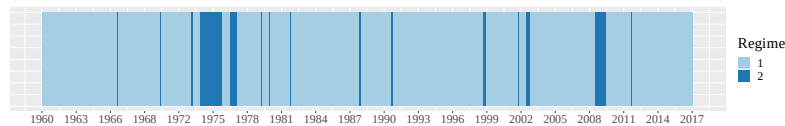
(b) Regimes based on PPI



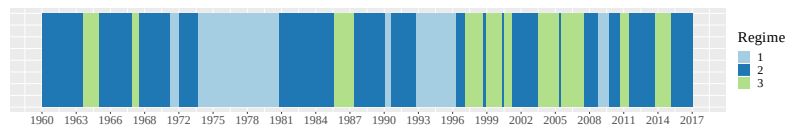
(c) Regimes based on changes in the bank deposit rate



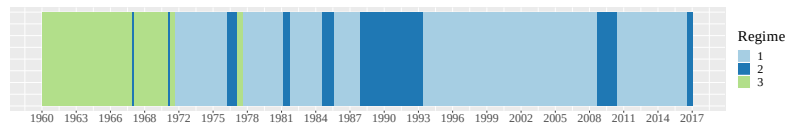
(d) Regimes based on changes in medium-term yields



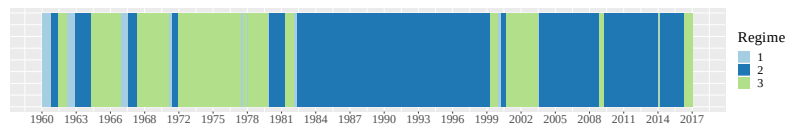
(e) Regimes based on stock returns



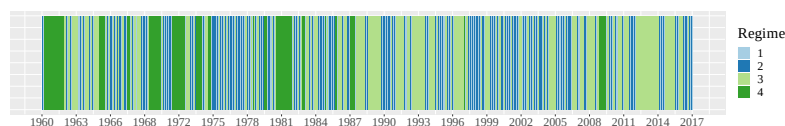
(f) Regimes based on changes in corporate bond spreads



(g) Regimes based on changes in the dollar/pound exchange rate



(h) Regimes based on changes in notes and coins in circulation



(i) Regimes based on changes in credit extension

Figure 4: Regimes based on univariate hidden Markov models

regime in March 2008, *before* the global financial crises in late-2008.

The MRCM for this model is 29.7%, suggesting some ambiguity in the classification. The regimes are, in general, quite persistent, with an RP of 87%.

Table 2: Industrial production regimes

Regime	Number of observations	Mean	Standard deviation
1	35	0.06	3.35
2	284	-0.02	0.46
3	365	0.01	0.81

4.1.2 PPI

In sharp contrast to industrial production, the model fitted seven regimes to PPI (Figure 4b). The descriptive statistics of the standardised PPI in each regime is shown in Table 3. The hyperinflation of the 1970s has been separated in regime 4, while the relatively benign inflationary environment of the 1990s and 2000s have been separated in regime 2. The persistence in the regimes were low, however, making it more difficult to use the regimes for forecasting and risk-management. The MRCM for this model is relatively high at 39.8%, indicating that there is significant ambiguity in the classification. Similarly, the model indicates low regime persistence, with an RP of only 55%.

Table 3: PPI regimes

Regime	Number of observations	Mean	Standard deviation
1	53	0.98	0.51
2	111	-0.72	0.61
3	191	-0.33	0.26
4	75	1.79	1.45
5	99	0.22	0.40
6	33	0.57	0.13
7	122	-0.69	0.00

4.1.3 Bank deposit rate

The model identified eight regimes in the short-term interest rates, as captured by the bank deposit rate (Figure 4c). However, in this case, there is more of a gradual change over time. In the initial part of the study period (1960-1980), rates were stable (regime 3), interspersed occasionally with relatively sharp positive and negative changes (regimes 1 and 4). Between 1980 and 2000, rates were relatively volatile, with sharp changes occurring more frequently (regimes 4, 6 and 7). In the remainder of the period, rates were relatively stable, with slightly higher volatility. The MRCM for this model was 19.65%, which is relatively low, suggesting that the regime classification was more clear. In this case, however, the regime persistence was very low at 42%.

Table 4: Bank deposit rate regimes

Regime	Number of observations	Mean	Standard deviation
1	24	-0.90	0.00
2	10	-1.81	0.00
3	294	0.00	0.00
4	64	1.71	2.26
5	128	0.00	0.05
6	48	-0.53	0.31
7	27	-1.80	0.74
8	89	0.04	0.20

4.1.4 Medium-term yield

In this case, the model identified only two regimes using medium-term yields (Figure 4d). Between 1960 and 1970, changes in yields tend to be marginally below average with relatively low volatility, while between 1970 and the mid-1990s, changes in yields were marginally above average with much more volatility, before reverting once again to the more stable regime 2. These two regimes can be visually recognised in Figure 1. The MRCM for this model was very low at 8.67%, indicating that the classification over time was fairly unambiguous. The regimes were also very persistent, with an overall RP of nearly

100%.

Table 5: Medium-term yield regimes

Regime	Number of observations	Mean	Standard deviation
1	276	0.02	1.44
2	408	-0.02	0.53

4.1.5 Stocks

The HMM also identified only two regimes based on stock returns (Figure 4e). The "equilibrium" regime is characterised by above-average returns with relatively low volatility, with the "crisis" regime characterised by sharply lower returns with very high volatility (Table 6). The notable periods of persistence in the crisis regime was in the 1974-1977 and again during the credit crisis between June 2008 and May 2009, with shorter periods in 1998 and 2002. The MRCM for this model is 28.17%. The model suggests that stock market regimes are relatively persistent, with an overall RP of 84%. This is mainly due to the equilibrium regime where there is a 94% probability to remain in this regime.

Table 6: Stock return regimes

Regime	Number of observations	Mean	Standard deviation
1	618	0.09	0.72
2	66	-0.88	2.18

4.1.6 Corporate bond spreads

Corporate bond spreads provides some measure of the market's perception of risk. In this case, the HMM identified three regimes (Figure 4f), which are largely based on the difference in volatility of spread changes (Table 7). During regime 3, bond spreads were relatively stable, while regime 1 is characterised by significantly more volatility in bond spreads. The MRCM for this model is relatively high at 39.2%, suggesting that there was some ambiguity in the classification. Based on this model, the regimes of corporate bond spreads are fairly persistent, with an overall RP of 91%.

Table 7: Regimes based on changes in corporate bond spreads

Regime	Number of observations	Mean	Standard deviation
1	156	-0.05	1.75
2	368	0.04	0.73
3	160	-0.03	0.22

4.1.7 Dollar/pound exchange rate

The model identified three regimes based on the dollar pound exchange rate (Figure 4g). The pound was pegged to the dollar as part of the Bretton Woods system up until August 1971, which is clearly identified in the model as regime 3. Regime 2 is characterised by sharp devaluation in the pound against the dollar with significant volatility, while regime 1 is characterised by moderate strengthening of the currency. The MRCM for this model is 22.8%. The regimes are quite persistent, with an RP of 96%.

Table 8: Regimes based on changes in the dollar/pound exchange rate

Regime	Number of observations	Mean	Standard deviation
1	414	0.09	0.84
2	129	-0.33	1.71
3	141	0.05	0.07

4.1.8 Notes and coins in circulation

In contrast to PPI, the model identified three regimes based on changes in notes and coins in circulation (Figure 4h). Regime 1 is characterised by sharp contraction in narrow money supply, with significant volatility. Regime 2 is associated with more stable contraction, whereas monetary control was much looser during regime 3 with strong growth in money supply, most notably during the period of hyper inflation in the 1970s; and during the aftermaths of the IT bust in 2000 and the credit crisis of 2008. More recently, the model switched to regime 3 in April 2016. The MRCM for this model was 30.1%. Based on this model, the regimes in money supply are relatively persistent, with an RP of 84%.

Table 9: Regimes based on changes in M1 money supply

Regime	Number of observations	Mean	Standard deviation
1	39	-0.34	3.11
2	412	-0.16	0.44
3	233	0.34	0.91

4.1.9 Credit extension

Finally, the regimes for credit extension is shown in Figure 4i. The model identified four regimes, with very low regime persistence. In this case, the MRCM was 48.5%, indicating that there was significant ambiguity in regime classification. The regime persistence based on this model is also very low, with an overall RP of only 43%. This is the lowest RP of the variables that we considered. While there seems to be differences between the statistics of the various identified regimes (Table 10), they can hardly be considered homogeneous regimes, due to their relative short duration.

Table 10: Regimes based on changes in Credit

Regime	Number of observations	Mean	Standard deviation
1	157	-0.69	0.36
2	169	0.60	0.35
3	250	0.00	0.26
4	108	0.06	2.14

After having examined individually each economic indicator, in the next section we consider the indicators together in order to identify broad macroeconomic regimes.

4.2 Multivariate HMM on economic indicators

From Figure 4 it is clear that each economic indicator has its own regimes, which sometimes (but not always) overlap with the regimes of other indicators. A multivariate hidden Markov model can be used to identify regimes based on multiple indicators, by making assumptions on the multivariate distributions within each regime.

We trained a multivariate HMM using the nine indicators and assumed that they are

multivariate Gaussian distributed within specific regimes. As above, we determined the number of regimes using the BIC. In this case, the HMM identified seven different regimes, shown in Figure 5. The statistics for each of the nine variables in each of the seven regimes appear in Table 11.

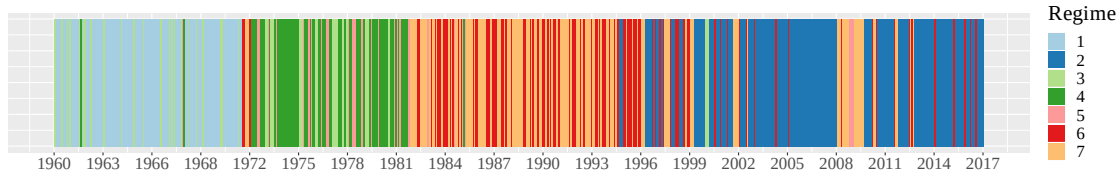


Figure 5: Multivariate regimes based on all macroeconomic indicators

Visually, there seems to be four major "equilibrium" regimes. The initial period until mid 1971 was dominated by regime 1, which is characterised by improving IP and negative inflation. Regime 4 in the second period between mid-1971 and the end of 1982 was less persistent, and characterised by declining IP, rising inflation, increasing bond yields and lower stock returns. The third period between 1983 and the first quarter of 1996 was dominated by regime 7, characterised by lower yields and interest rates. Finally the last period from 1996 until the end of the study period was dominated by regime 2, characterised by lower inflation.

Each of these sub-periods contains at most another three regimes. In the first part, regime 3 (with lower interest rates and higher stock returns) occurred occasionally with low persistence, while regimes 3 and 5 often interrupted the second equilibrium period. Regime 6 (higher stock returns, lower money supply) upset the equilibrium in both the last two periods. Based on this model, the model seldom if ever returns to a previous equilibrium regime - the only exception is the return to regime 7 during the credit crisis of 2008. This in itself suggests a Markovian property: historical information is not that important when modelling the current economic environment.

The transition matrix for the above HMM appear in Table 12. Not surprisingly, the regime persistence of the equilibrium regimes are quite high, but the short persistence of the interrupting regimes could pose a problem for inference. The regime persistence of this model (from (9)) is 55%, which indicate relatively low regime persistence. The MRCM for this model is 6.7%, which indicates that there was little ambiguity in the regime

Table 11: Statistics of economic indicators in regimes based on economic indicators

	Regime	1	2	3	4	5	6	7
	# of obs	111	179	48	77	32	105	132
IP	mean	0.16	0.01	0.09	-0.22	0.05	0.00	-0.07
	sd	1.31	0.46	1.28	1.63	1.22	0.75	0.71
PPI	mean	-0.15	-0.48	0.39	1.39	0.55	-0.18	-0.16
	sd	0.72	0.48	1.55	1.28	1.16	0.64	0.60
Bank_deposit	mean	0.00	0.02	-0.45	0.00	1.83	0.00	-0.32
	sd	0.00	0.13	1.65	0.00	3.14	0.00	0.94
MT_yield	mean	0.06	-0.01	-0.71	0.50	1.08	-0.09	-0.26
	sd	0.49	0.49	1.61	1.15	1.51	0.75	1.14
Stocks	mean	-0.02	-0.02	0.52	-0.27	-0.60	0.25	-0.04
	sd	0.83	0.60	1.60	1.21	1.31	0.78	1.10
CB_spreads	mean	0.09	0.01	-0.05	-0.03	0.11	-0.14	0.04
	sd	0.59	0.43	1.32	1.61	1.75	0.84	1.11
Dollar_pound	mean	0.06	0.10	0.20	-0.19	-0.81	0.05	0.02
	sd	0.07	0.83	0.39	1.19	1.39	0.99	1.39
M1	mean	-0.09	-0.08	-0.03	0.62	0.39	-0.19	-0.11
	sd	1.47	0.58	2.06	0.95	0.63	0.49	0.55
Credit	mean	0.02	0.04	0.04	-0.16	0.09	-0.01	-0.01
	sd	1.22	0.54	0.97	1.66	1.81	0.64	0.71

classification.

Table 12: Transition matrix of Hidden Markov Model based on untransformed economic indicators

Regime	1	2	3	4	5	6	7
1	0.81	0.00	0.18	0.00	0.00	0.01	0.00
2	0.00	0.85	0.01	0.00	0.00	0.12	0.02
3	0.41	0.03	0.33	0.14	0.09	0.00	0.00
4	0.02	0.00	0.12	0.66	0.20	0.00	0.00
5	0.00	0.00	0.00	0.55	0.29	0.00	0.16
6	0.00	0.18	0.00	0.01	0.02	0.34	0.45
7	0.00	0.05	0.00	0.00	0.02	0.35	0.58

The regime shifting process can also be graphically represented through a directional graph (Figure 6). Each circle depicts a regime and the connecting arches represents the transition mechanism. The thickness of the arches signify the strength of the transition probability while the circumference of the circle depict the number of observations of that particular regime. The most likely path from each regime can then be followed. For instance, from regime 1 it is unlikely to shift directly to regimes 2, 5, 6 and 7, while it is most likely to remain in regime 1. Similarly, the regime persistence of regime 6 is fairly weak, with a shift to regime 7 more likely.

4.2.1 Model sensitivity

From the regime changes on individual variables, the bank deposit rate (see Figure 4c) also showed a gradual evolution to different regimes. If this variable on its own is driving the change in the multivariate model, one could only consider the regime shifts in that variable. However, by removing the bank deposit rate from the dataset, the different equilibrium regimes still appear, albeit with even less regime persistence (See Figure 11 and Table 15 in Appendix A).

Similarly, one could argue that the second-order differenced Credit variable introduces significant noise to the model, without adding any additional clarity on regime changes.

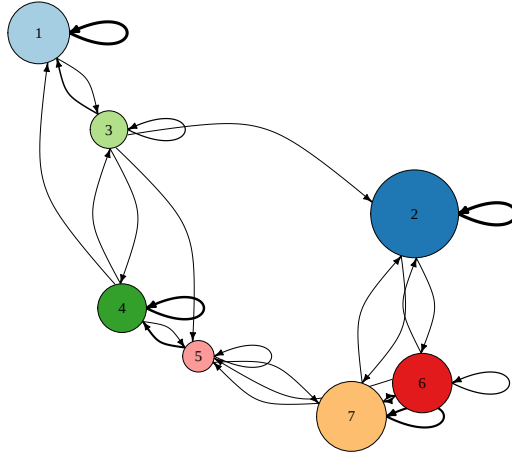


Figure 6: Graph representation of multivariate regimes

Removing this variable from the dataset, however, resulted in a model with even less persistence (See Figure 12 and Table 16 in Appendix A).

The two methodologies used so far have showed how regimes change, but we observe a high level of complexity which is detrimental in testing our objectives: 1) non-linearity of the broad economic system and 2) the identification of the latent influences which cause the shifts. In the next section, we test our two-step process.

5 Hidden Markov models with ICA preprocessing

We now use our two-step process to first extract independent components from the economic indicators and then use those in regime detection, and compare the results with those obtained in Section 4.2.

5.1 Independent component extraction using SOBI

The independent components extracted using SOBI appear in Figure 7, and their correlations with the original variables appear in Figure 8.

The correlations between the independent components and the original variables provide an indication of the nature of each influence. While each of the independent components is highly correlated with an original observed series, there are secondary factors that

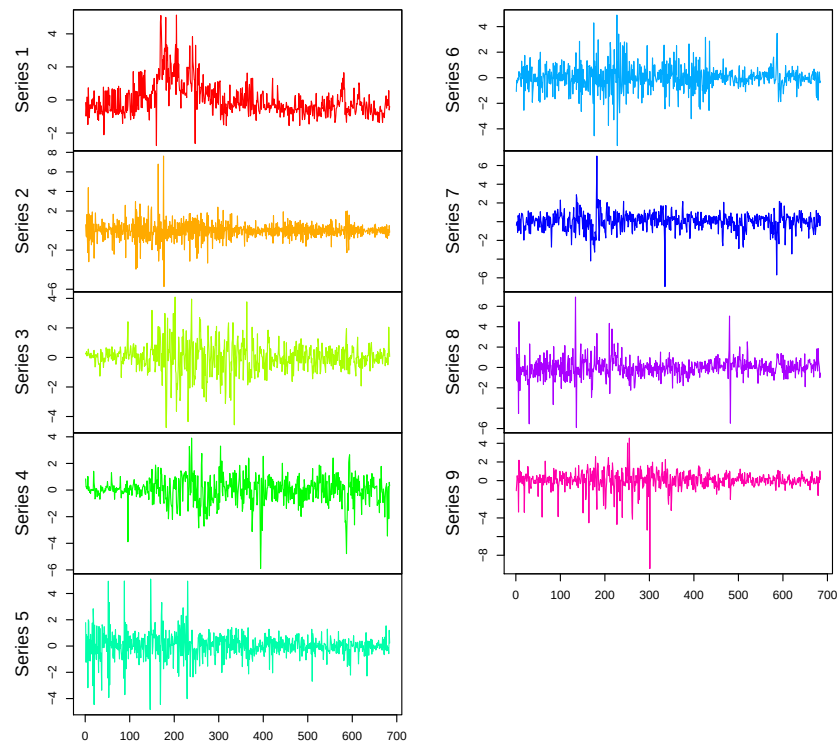


Figure 7: Timeseries of independent components

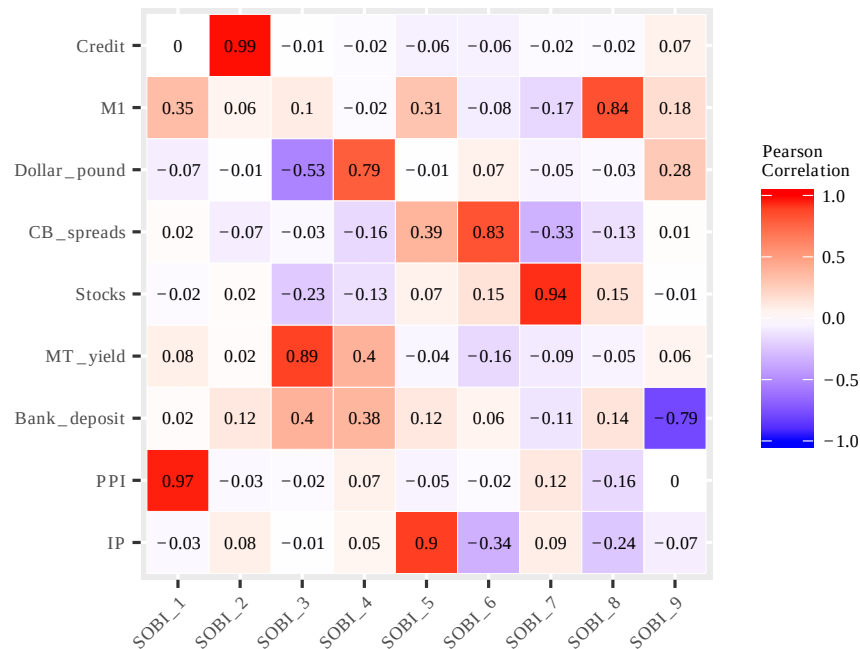


Figure 8: Correlations of independent components with original variables

shed light on the cross-influences. In essence, the independent components capture all the original impacts, but with the added benefit that the components are now all independent from each other, therefore providing a "purer", specific economic influence.

SOBI_1 captures inflationary influences, with PPI and narrow money supply (M1) contributing to this component. SOBI_2 encapsulates the influence of credit extension, while SOBI_3 represents the impact of medium-term yields, but removing the cross-impact of the currency and stock returns.

Similarly, SOBI_4 depicts the effect of the exchange rate, but also extracts the impact of the bank deposit rate and medium-term yields on the exchange rate. SOBI_5 captures industrial production, with contributions from money supply and corporate bond spreads. Chan-Lau and Ivaschenko (2002) has previously identified the influence of corporate bond spreads on industrial production in the United States. SOBI_6 extracts the influence of corporate bond spreads, without the impact on industrial production and medium-term yields.

SOBI_7 extracts mainly the impact of stock returns, but removes the impact from corpo-

rate bond spreads, money supply and the bank deposit rate. SOBI.8 extracts the influence of narrow money supply, without the influence of industrial production and inflation. Finally, SOBI.9 extracts the short-term interest rate effect, without the counteraction of the exchange rate and narrow money supply.

We can conclude thus by observing that each SOBI component is influenced by no more than three economic indicators, and in each case there is a dominant influence. This could facilitate with interpreting the SOBI components.

5.2 Multivariate HMM on independent components

The modelled regimes from a HMM using independent components as input appear in Figure 9, with the statistics of the economic indicators in each regime appearing in Table 13. The statistics of the independent components in each regime appear in Table 17 in the Appendix.

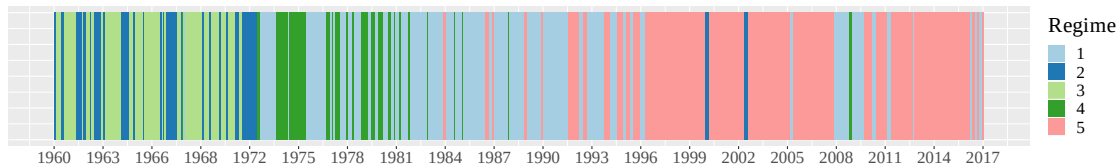


Figure 9: Multivariate regimes based on independent components

The HMM identified five different regimes in total, and visually, the classification of regimes appears more clear than that based on the untransformed economic indicators.

There are three main "equilibrium" regimes over the study period. Similar to the model based on untransformed variables (see Figure 5), the first equilibrium regime lasted until mid-1971. This coincides with the stable exchange rate due to the Bretton Woods system. In addition, volatility across most macroeconomic indicators were low, with above average growth in industrial production and below average inflation. Changes in medium-term yields were above average, while changes in the bank deposit rate were marginally below average. Stock returns were relatively stable and marginally above the long-term average. Changes in narrow money supply were below average. This regime occurred only in the earlier part of the study period and was often interrupted by regime 2. If we compare our first analysis (Figure 5 - Model 1) with this result, we observe that the first

Table 13: Statistics of economic indicators in regimes based on independent components

	Regime	1	2	3	4	5
	# of obs	227	61	92	58	246
IP	mean	0.01	-0.10	0.15	-0.16	-0.00
	sd	0.79	2.15	0.83	1.65	0.42
PPI	mean	0.27	-0.20	-0.16	1.64	-0.53
	sd	0.80	0.59	0.73	1.77	0.42
Bank_deposit	mean	-0.20	0.06	-0.06	0.80	0.01
	sd	0.82	1.03	0.23	2.65	0.23
MT_yield	mean	-0.08	-0.09	0.17	0.47	-0.07
	sd	1.18	0.68	0.37	2.04	0.54
Stocks	mean	0.11	-0.17	0.04	-0.33	0.01
	sd	0.90	0.89	0.82	2.17	0.68
CB_spreads	mean	-0.06	-0.07	0.05	0.27	-0.01
	sd	1.09	1.00	0.51	2.17	0.48
Dollar_pound	mean	-0.11	0.16	0.05	-0.24	0.10
	sd	1.34	0.39	0.07	1.34	0.80
M1	mean	0.04	-0.20	-0.17	0.67	-0.08
	sd	0.72	2.36	0.87	1.04	0.53
Credit	mean	-0.02	0.24	-0.11	-0.09	0.02
	sd	0.78	1.20	1.23	2.07	0.54

equilibrium regime in Model 1 was interrupted often and with low persistence by regime 3. Instead in this case, the interrupting regime 2 shows more persistence, with a 68% probability of remaining in the same state and a 29% probability of returning to regime 3. Regime 2 is characterised by higher volatility across most economic indicators, in particular industrial production and narrow money supply. The occurrence of this regime more or less aligns with the UK recession towards the end of 1961, and the lower-growth regime of industrial production (see Figure 4a). This regime provided the transition between the two equilibrium regimes towards the end of 1971. It occurred predominantly in this period, although was found to be the most likely regime for short periods towards the end of 1999 and in mid-2002.

The second equilibrium regime features mainly regime 1, which occurred between mid-1972 and mid-1994, and initially was interrupted by regime 4 and later by regime 5. The equilibrium regime was characterised by above-average increases in inflation, above-average stock returns, a weaker currency and declining bank deposit rates. In the earlier part of this period, this equilibrium phase was interrupted by periods of significantly higher inflation, low stock returns and higher interest rates (regime 4). The volatilities of most economic indicators during this "crisis" regime were also much higher than in other periods. This period includes the oil crises of 1973 and 1979 and the periods of stagflation it followed. It is interesting to note that based on this model, the equilibrium regime does not correspond with the Great Moderation of Stock and Watson (2003), which started in 1985. The start of the Great Moderation does coincide with the end of the frequent occurrence of regime 4, however, and this second equilibrium regime was interrupted by an even lower-volatility regime (regime 5), which grew more persistent in the late 1990s until eventually becoming the final equilibrium regime.

The final equilibrium period from 1996 onwards was characterised by low volatility across most of the economic indicators that we considered in this analysis. According to this model, this new era of moderation was interrupted briefly by regime 2 in the late 1999-early 2000, coinciding with recessionary conditions elsewhere in the world, but not in the UK. Regime 2 once again interrupted the equilibrium in 2002, when growth in the UK stagnated briefly. The equilibrium also changed to the slightly more volatile regime 1 (which dominated the 1980s) at the end of 2007, before a brief return to the crisis regime

(regime 4) during the worst part of the credit crisis in 2008. The economy finally return to the equilibrium of regime 5 in September 2009.

We can observe from the diagonal entries of the transition matrix for this model (Table 14) that the regimes are persistent and based on (9), the regime persistence of the model is 75%, considerably higher than that of the model based on the untransformed indicators. This is an important outcome of the methodology because it will be beneficial to support or develop strategies based on economic regimes, such as asset allocation. Nonetheless, the MRCM of this model is somewhat higher at 18.8%, indicating that there is some ambiguity in classifying observations into regimes. However, in our opinion, the reduced complexity of the model, coupled with its higher regime persistence outweighs the deterioration in the MRCM, relative to that of the model based on the untransformed variables.

Table 14: Transition matrix of Hidden Markov Model based on independent components

	1	2	3	4	5
1	0.80	0.00	0.01	0.10	0.10
2	0.00	0.63	0.29	0.03	0.05
3	0.01	0.22	0.77	0.00	0.00
4	0.35	0.00	0.00	0.65	0.00
5	0.09	0.01	0.00	0.00	0.90

Furthermore, the economic equilibrium regimes seem to evolve over time, with the stable regimes getting longer and more persistent. While the initial equilibrium regimes were often interrupted by short-lived alternative regimes, the last regime was very persistent, perhaps suggesting a new era of moderation. This confirms the model of Goodwin (1951), who argued that the model economy evolves smoothly with multiple equilibria.

The evolution of regimes through multiple equilibria also suggest that it is unlikely that previous equilibrium regimes would be revisited: once the equilibrium has shifted, it is unlikely to recur. This has implications for decision processes dependent on equilibrium assumptions. From a classical statistical perspective, more data is normally preferable to obtain a better understanding of the true distribution of variables. However, this analysis has shown that in economics, historical data may no longer be applicable if the equilibrium regime has shifted. For instance making asset allocation decisions based on the risk-return

characteristics of assets over the entire 56 year history is not advisable, since the conditions that existed in the 1960s are unlikely to recur.

The outcomes obtained in this implementation of the proposed methodology is clear when considering the directional graph of the regime shifting process, as shown in Figure 10. While the probability of remaining in each of the three equilibrium regimes (regimes 1, 3 and 5) is high, the graph indicates that the process is more likely to move to one of the equilibrium regimes than the two crisis regimes.

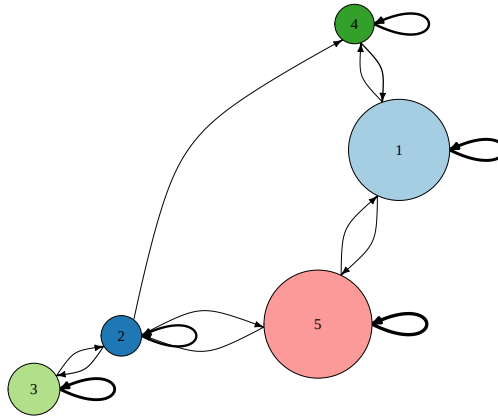


Figure 10: Graph representation of multivariate regimes based on independent components

6 Conclusions

In this paper, we have examined two critical aspects of macroeconomic modelling related to regime changes: non-linearity of the economic system and mixtures of independent economic influences. In doing so we have developed a new method based on the combination of hidden Markov Models and independent component analysis.

We illustrated the model using nine economic indicators from the UK. The individual time series have different regimes which seldom overlap and few indicators have more than three different regimes. The results show that preprocessing the time series first to extract independent components helps in reducing the noise in the hidden Markov Model, enabling fewer regimes to be recognised with greater regime persistence. This last finding could

provide useful information when regime-specific decisions has to be made, such as asset allocation.

Several areas of future research can be explored: while the preprocessing of data using independent component analysis provided clearer, more persistent regimes, it is unclear whether this is always the case. Furthermore, more research is needed on how to apply this research within an asset allocation and portfolio construction framework and we are currently pursuing this line of research. Thirdly, in this paper we considered the in-sample *identification* of regimes, but more research should be done on the *prediction* of future regimes. Finally, as suggested in Zhu et al. (2017), the presence of longer memory in economic regimes could be investigated through higher-order hidden Markov models.

A Alternative models

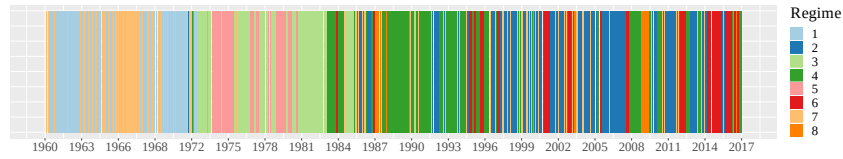


Figure 11: Multivariate regimes excluding bank deposit rates

Table 15: Transition matrix of Hidden Markov Model based on economic indicators excluding bank deposit rates

Regime	1	2	3	4	5	6	7	8
1	0.61	0.00	0.02	0.00	0.00	0.00	0.37	0.00
2	0.00	0.66	0.00	0.04	0.00	0.06	0.23	0.01
3	0.00	0.00	0.79	0.03	0.13	0.00	0.05	0.01
4	0.00	0.02	0.02	0.72	0.00	0.07	0.11	0.06
5	0.00	0.00	0.24	0.00	0.76	0.00	0.00	0.00
6	0.00	0.12	0.00	0.16	0.00	0.70	0.00	0.02
7	0.22	0.25	0.03	0.15	0.00	0.00	0.35	0.00
8	0.06	0.12	0.00	0.13	0.00	0.09	0.14	0.47

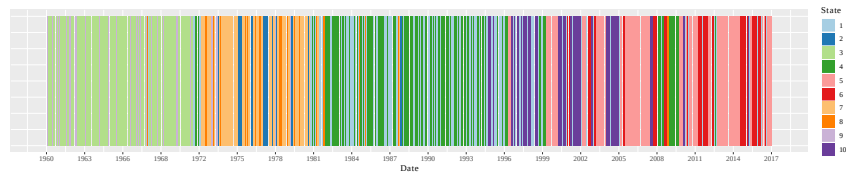


Figure 12: Multivariate regimes excluding credit

Table 16: Transition matrix of Hidden Markov Model based on economic indicators excluding credit

Regime	1	2	3	4	5	6	7	8	9	10
1	0.34	0.00	0.01	0.37	0.08	0.05	0.01	0.05	0.00	0.10
2	0.00	0.53	0.00	0.05	0.00	0.00	0.42	0.00	0.00	0.00
3	0.01	0.00	0.81	0.00	0.00	0.00	0.00	0.00	0.18	0.00
4	0.36	0.00	0.00	0.60	0.03	0.01	0.00	0.00	0.00	0.00
5	0.01	0.00	0.00	0.00	0.86	0.07	0.00	0.00	0.00	0.06
6	0.14	0.00	0.00	0.02	0.09	0.66	0.00	0.02	0.00	0.07
7	0.03	0.09	0.00	0.00	0.00	0.00	0.68	0.15	0.06	0.00
8	0.08	0.00	0.00	0.13	0.00	0.00	0.54	0.25	0.00	0.00
9	0.00	0.10	0.67	0.00	0.00	0.00	0.00	0.07	0.16	0.00
10	0.33	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.65

B Statistics of independent components in regimes

Table 17: Statistics of independent components in regimes

	Regime	1	2	3	4	5
	# of obs	227	61	92	58	246
SOBI.1	mean	0.27	-0.21	-0.21	1.73	-0.52
(Inflation)	sd	0.81	0.71	0.74	1.63	0.39
SOBI.2	mean	-0.03	0.23	-0.11	-0.02	0.02
(Credit)	sd	0.78	1.20	1.23	2.07	0.54
SOBI.3	mean	-0.05	-0.17	0.16	0.34	-0.05
(Yields)	sd	1.21	0.62	0.33	1.99	0.58
SOBI.4	mean	-0.18	0.10	0.10	0.25	0.04
(Exchange rate)	sd	1.28	0.60	0.20	1.52	0.77
SOBI.5	mean	-0.01	-0.19	0.13	0.01	0.01
(Growth)	sd	0.81	2.12	0.82	1.68	0.41
SOBI.6	mean	-0.08	-0.05	0.04	0.27	0.01
(Spreads)	sd	1.11	1.19	0.64	2.01	0.44
SOBI.7	mean	0.11	-0.18	0.11	-0.26	-0.03
(Equities)	sd	0.87	1.03	0.82	2.06	0.74
SOBI.8	mean	-0.06	-0.10	-0.18	0.15	0.12
(Money supply)	sd	0.77	2.18	0.86	1.37	0.58
SOBI.9	mean	0.11	-0.10	0.16	-0.60	0.01
(Short-term rates)	sd	0.88	1.20	0.34	2.43	0.44

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